

Math 135

Please give me feedback! I am at the first row.

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1 Admin

1.1 General Information

1.1.1 Contact

For mathematics questions: `abeltaos@uwaterloo.ca`

For course logistics: `math135@uwaterloo.ca`

1.1.2 Teaching Style

The LEARN website contains several useful resources. In particular:

- (i) The icons at the top of the page provide quick access to important resources, including the course schedule.
- (ii) Crowdmark is used for submitting assignments.

1.1.3 Graded Assignments

There are three assignments each week.

1.1.4 Late Submission Policy

Late submissions are accepted with a penalty of 1% per minute for the entire assignment.

1.1.5 Office Hours

Monday and Wednesday 2:00 to 3:30 in (MC 6503) or by appointment (email the instructor to arrange).

2 Introduction to the Language of Mathematics

2.1 Introduction to Proof

Definition 2.1.1 (Proof)

A proof is a logically valid argument that establishes the truth of a mathematical statement using definitions, axioms, and previously proved results.

Proposition 2.1.2

For every positive integer n , the number $n^2 + 1$ is not a perfect square.

Proof

Let n be a positive integer. The consecutive perfect squares surrounding $n^2 + 1$ are n^2 and $(n + 1)^2$. Indeed,

$$n^2 < n^2 + 1 < n^2 + 2n + 1 = (n + 1)^2,$$

where the second inequality follows from $n > 0$.

Therefore, $n^2 + 1$ lies strictly between two consecutive perfect squares, so it cannot itself be a perfect square. ■

2.2 Sets

Definition 2.2.1 (Set)

A **set** is a well-defined, unordered collection of distinct objects. Each object that appears in this collection is called an **element** of the set.

For a finite set, we can denote it by listing its elements, separated by commas, surrounding the entire list by braces (or by $\{\}$).

Example 2.2.2

The set containing the elements 2, 7, and -3 can be denoted by

$$\{7, -3, 2\}.$$

Example 2.2.3

Here are some examples:

- $\{2, 4, 6, 8\}$
- $\{1, 2, \{1, 2, 3\}\}$
- $\{a, b, c, d\}$
- $\mathbb{N} = \{1, 2, 3, \dots\}$
- $\mathbb{Z} = \{\dots, -1, 0, 1, \dots\}$
- $\mathbb{Q} = \{\frac{a}{b} : a, b \in \mathbb{Z}, b \neq 0\}$

Fact: $|\mathbb{N}| = \aleph_0$, $|\mathbb{R}| = \aleph_1$ $\aleph_1 = 2^{\aleph_0}$.

Definition 2.2.4

The **parity** of an integer is **even** if the integer is a multiple of 2, and **odd** otherwise.

2.3 Statement and Negation

Definition 2.3.1 (Statement)

A **statement** is a sentence that has a definite state of being either true or false.

Definition 2.3.2 (Negation)

Suppose that A is a statement. Then the negation of A , denoted by $\neg A$, is the statement asserting the opposite truth value to A . That is, $\neg A$ is false when A is true, and $\neg A$ is true when A is false.

2.4 Quantifiers and Quantified Statements

2.4.1 Universal Quantifier

$\forall$ means "for all", and it is called the universal quantifier.

Example 2.4.1

$$\forall x \in \mathbb{Z}, x \geq 5, 2^n > n^2.$$

The negation of universal quantifier.

$$\neg(\forall x \in S, P(x)) \equiv (\exists x \in S, \neg P(x))$$

2.4.2 Existential quantifier

$\exists$ means "there exists", and it is called the existential quantifier.

Example 2.4.2

$$\exists a \in \mathbb{Z}, a^2 + 29a + 209 \leq 0.$$

The negation of an existentially quantifier statement (such as $\exists x \in S, P(x)$) is:

$$\neg(\exists x \in S, P(x)) \equiv (\forall x \in S, \neg P(x)).$$

Remark

A quantified statement contains four parts:

- variable;
- quantifier;
- domain;
- an open sentence involving the variable (that is either true or false whenever a value of the variable is chosen from the domain).

2.4.3 Nested Quantifier

Consier this statement

$$\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, x > y.$$

That is a true statement. (Given $x \in \mathbb{R}$, you could provide $y = x - 1$ to prove the existential statement)

Remark

Order matters! Consider the statement: $\exists y \in \mathbb{R}, \forall x \in \mathbb{R}, x > y$. This says there is a real number y which is smaller than every real number! False!!!

To negative a nested quantifier statement, think about it in pieces. For example:

$$\neg (\forall x \in S, \exists y \in T, P(x, y)) \equiv \exists x \in S, \forall y \in T, \neg P(x, y).$$

3 Logical Analysis of Mathematical Statements

3.1 Truth Table and Negation

Definition 3.1.1

The truth value for "not A ", written symbolically as $\neg A$, is defined by the truth table given below.

A	$\neg A$
T	F
F	T

We also refer to $\neg A$ as the negation of A .

We can use the truth table to prove that an argument is equal to another argument.

Example 3.1.2

Given the truth table for $A, \neg A, \neg(\neg A)$,

A	$\neg A$	$\neg(\neg A)$
T	F	T
F	T	F

This truth table serves as a proof that $A \equiv \neg(\neg A)$.

Remark

" A " is considered a statement variable.

3.2 Conjunction and Disjunction

Definition 3.2.1 (Compound statement)

A compound statement is a statement composed of several individual statements, each of which is called a component statement.

Definition 3.2.2

The truth value for $A \wedge B$, is defined by the truth table

A	B	$A \wedge B$
T	T	T
T	F	F
F	T	F
F	F	F

We also refer to $A \wedge B$ as the conjunction of A and B .

If we have three component statement, we need 2^3 rows to cover all the possibility.

Example 3.2.3

The compound statement

$$(2 + 3 = 6) \wedge (\pi + 2 > 5)$$

is false, since the statement $2 + 3 = 6$ is false.

Definition 3.2.4

The truth value for A or B , written as $A \vee B$, is defined by the truth table

A	B	$A \vee B$
T	T	T
T	F	T
F	T	T
F	F	F

We also refer to $A \vee B$ as the **disjunction** of A and B .

Remark

The disjunction $A \vee B$ is the *inclusive* or: it is true whenever at least one of A , B is true, including the case where both are true. This differs from the *exclusive* or (xor), written $A \oplus B$, which is true when exactly one of A , B is true and false when both are true. Thus $A \vee B$ and $A \oplus B$ agree in every case except $A = B = \text{T}$, where $A \vee B$ is true but $A \oplus B$ is false.

Example 3.2.5

The Compound statement:

$$(2 + 3 = 6) \vee (\pi + 2 \geq 5),$$

is true since one of the statements is true.

Example 3.2.6

Let us use a truth table to prove that

$$\neg(A \vee B) \equiv (\neg A \wedge \neg B)$$

A	B	$A \vee B$	$\neg(A \vee B)$	$\neg A$	$\neg B$	$\neg A \wedge \neg B$
T	T	T	F	F	F	F
T	F	T	F	F	T	F
F	T	T	F	T	F	F
F	F	F	T	T	T	T

A similar equivalence is

$$\neg(A \wedge B) \equiv ((\neg A) \vee (\neg B)).$$

These things together are called DeMorgan's Law.

Theorem 3.2.7 (DeMorgan's Law)

For statement variable A and B , we have

$$\neg(A \vee B) \equiv (\neg A \wedge \neg B)$$

$$\neg(A \wedge B) \equiv (\neg A \vee \neg B)$$

Proposition 3.2.8

For statements A, B, C , the following rules hold for the logical operations $\wedge$ and $\vee$:

Commutative Laws

$$A \vee B \equiv B \vee A$$

$$B \wedge A \equiv A \wedge B$$

Associative Laws

$$A \vee (B \vee C) \equiv (A \vee B) \vee C$$

$$A \wedge (B \wedge C) \equiv (A \wedge B) \wedge C$$

Distributive Laws

$$A \vee (B \wedge C) \equiv (A \vee B) \wedge (A \vee C)$$

$$A \wedge (B \vee C) \equiv (A \wedge B) \vee (A \wedge C)$$

3.3 Implication

Definition 3.3.1

The truth value for " A implies B ", written symbolically as $A \implies B$, is defined by the truth table

A	B	$A \implies B$
T	T	T
T	F	F
F	T	T
F	F	T

We also refer to $A \implies B$ as an implication.

Just for fun. Instructor does not talk about it.

$$A \implies B \equiv \neg(A \wedge \neg B) \equiv \neg A \vee B.$$

Hypothesis: Assume to be true: A

Conclusion: B

Example 3.3.2

$$(\forall x \in \mathbb{R}) [(x > 2) \implies (x^2 > 4)]$$

Remark

Interesting.

$$(A \vee B) \implies C \equiv (A \implies C) \wedge (B \implies C)$$

3.4 Converse and Contrapositive

3.4.1 Converse

The $B \implies A$ is called the converse of $A \implies B$.

A	B	$A \implies B$	$B \implies A$
T	T	T	T
T	F	F	T
F	T	T	F
F	F	T	T

Remark

$(A \implies B) \not\equiv (B \implies A)$ (not logical equivalent)

3.4.2 Contrapositive

The implication $(\neg B) \implies (\neg A)$ is called the contrapositive of $A \implies B$.

A	B	$\neg A$	$\neg B$	$A \implies B$	$(\neg B) \implies (\neg A)$
T	T	F	F	T	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	T	T

Remark

$(A \implies B) \equiv ((\neg B) \implies (\neg A))$

An implication is also logically equivalent to another type of compound statement, as shown in the following truth table:

A	B	$\neg A$	$A \implies B$	$\neg A \vee B$
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

Remark

The negation of an implication is not an implication.

3.4.3 If and Only If

Definition 3.4.1

The truth value for "A if and only if B", written symbolically as $A \iff B$, is defined by the truth table.

A	B	$A \iff B$
T	T	T
T	F	F
F	T	F
F	F	T

4 Proving Mathematical Statement

4.1 Proving Universally Quantified Statements

A **proposition** is a true statement that has been proved to be true. A **theorem** is a particularly significant proposition. A **lemma** is a subsidiary proposition. A **corollary** is a proposition that follows almost immediately from a theorem.

Example 4.1.1

$$C^2 = A^2 + B^2 - 2AB \cos \theta$$

Let's do our first prove.

Proposition 4.1.2

$$\forall x \in \mathbb{R}, x^2 + 1 \geq 2x.$$

Proof

Let x be an arbitrary real number. Therefore, $x - 1$ must also be a real number, and hence

$$(x - 1)^2 \geq 0.$$

Expanding the terms on the left side. We get

$$x^2 - 2x + 1 \geq 0.$$

Adding $2x$ to both sides yields

$$x^2 + 1 \geq 2x.$$

■

Proposition 4.1.3

Prove that

$$\forall x, y \in \mathbb{R}, x^4 + x^2y + y^2 \geq 5x^2y - 3y^2.$$

Proof

Let x and y be arbitrary real numbers. Then $x^2 - 2y$ is also a real number. So

$$(x^2 - 2y)^2 \geq 0.$$

Expanding, we get

$$x^4 - 4x^2y + 4y^2 \geq 0.$$

Adding $5x^2y$ to both sides, we have

$$x^4 + x^2y + 4y^2 \geq 5x^2y.$$

Now, subtract $3y^2$ from both sides:

$$x^4 + x^2y + y^2 \geq 5x^2y - 3y^2.$$

■

Exercises

Proposition 4.1.4

Prove

$$\forall \theta \in \mathbb{R}, \sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$$

Prove that

$$\forall x, y \in \mathbb{R}, \max(x, y) = \frac{x + y + |x - y|}{2}.$$

Proof

Case 1: If $x \geq y$, then $\max(x, y) = x$, and $x - y \geq 0$, so $|x - y| = x - y$.

Then

$$\frac{x + y + |x - y|}{2} = \frac{x + y + x - y}{2} = \frac{2x}{2} = x = \max(x, y)$$

So the claim holds when $x \geq y$.

Case 2: If $x < y$, then $\max(x, y) = y$, and $x - y < 0$. So

$$|x - y| = -(x - y) = y - x.$$

Then

$$\frac{x + y + |x - y|}{2} = \frac{x + y + y - x}{2} = \frac{2y}{2} = y = \max(x, y).$$

■

Example 4.1.5

Here is an example:

Proposition 4.1.6

Disprove:

$$\forall x \in \mathbb{R}, (x^2 - 1)^2 > 0.$$

Proof

Let $x = 1$, Then

$$(x^2 - 1)^2 = (1 - 1)^2 = 0 \not> 0.$$

Therefore, it is not the case.

■

Exercise

Proposition 4.1.7

Prove $\forall x \in \mathbb{R}, |x - 3| + 2|x + 2| \geq 5$

Definition 4.1.8

$$(a \mid b) \iff (\exists k, b = ak)$$

4.2 Proof by Contrapositive, Elimination

Definition 4.2.1 (Proof by Contrapositive)

Since $(A \implies B) \equiv (\neg B \implies \neg A)$, we can prove $(A \implies B)$ by proving that $(\neg B \implies \neg A)$. This method is useful when a direct proof of $A \implies B$ is awkward, but assuming $\neg B$ gives a concrete hypothesis that is easier to work with.

Example 4.2.2

Here is an example of contrapositive proof.

Proposition 4.2.3

For all integer x , if $4 \nmid x$ then $36 \nmid 3x$.

Proof

The contrapositive is

$$(36 \mid 3x) \implies (4 \mid x).$$

Let x be arbitrary integer. Assume that $36 \mid 3x$. Thus, there exists an integer k such that $3x = 36k$.

Thus

$$x = 12k = 4 \cdot (3k) \text{ and } 3k \in \mathbb{Z}.$$

Therefore, by the contrapositive method,

$$(\forall x \in \mathbb{Z}) [4 \nmid x \implies 36 \nmid 3x].$$

■

Example 4.2.4

Here is another example.

Proposition 4.2.5

$$(\forall a, b \in \mathbb{R}) [(ab \in \mathbb{R} \setminus \mathbb{Q}) \implies (a \in \mathbb{R} \setminus \mathbb{Q} \vee b \in \mathbb{R} \setminus \mathbb{Q})]$$

Proof

The contrapositive is $\forall a, b \in \mathbb{R}$, if $a \in \mathbb{Q}$ and $b \in \mathbb{Q}$, then $ab \in \mathbb{Q}$.

Let a, b be arbitrary in $\mathbb{R}$. Assume that a, b are both rational.

Then $\exists p, q, r, s \in \mathbb{Z}$ such that

$$a = \frac{p}{q}$$

and

$$b = \frac{r}{s}$$

and

$$q \neq 0$$

and

$$s \neq 0$$

.

Thus,

$$a \cdot b = \frac{pr}{qs} \quad \text{and} \quad qs \neq 0.$$

where $pr, qs \in \mathbb{Z}$.

By the contrapositive

$$\forall a, b \in \mathbb{R}, (ab \notin \mathbb{Q}) \implies (a \notin \mathbb{Q} \vee b \notin \mathbb{Q}).$$

■

Definition 4.2.6 (Elimination Method)

Let A, B, C be statement variables.

$$(A \implies (B \vee C)) \equiv ((A \wedge \neg B) \implies C)$$

This is useful when the conclusion is a disjunction $B \vee C$: instead of proving the disjunction directly, we assume one disjunct fails ($\neg B$) and show the other (C) must then hold.

You can prove this, or alternative

$$(A \wedge \neg C) \implies B.$$

Example 4.2.7

Here is an example:

Proposition 4.2.8

Prove that for all $x \in \mathbb{R}$, if $|2x - 6| = 4$, then $x > 3$ or $x = 1$.

Proof

The elimination form is

$$\forall x \in \mathbb{R}, (|2x - 6| = 4) \wedge (x \leq 3) \implies x = 1.$$

Let x be an arbitrary real number. Assume $|2x - 6| = 4$ and $x \leq 3$, and prove $x = 1$.

Since $|2x - 6| = 4$, thus

$$2x - 6 = \pm 4.$$

So $2x = 10$ or $2x = 2$, meaning

$$x = 5 \quad \text{or} \quad x = 1.$$

But since $x \leq 3$, so

$$x = 1.$$

■

Proposition 4.2.9

$$\forall a, b, c \in \mathbb{Z}. (\forall x \in \mathbb{Z}, a \mid (bx + c)) \implies (a \mid (c - 2b)).$$

Proof

Let a, b, c be arbitrary integers. Assume for all $x \in \mathbb{Z}, a \mid (bx + c)$.

Since $a \mid (bx + c)$ is true for all integers x .

Remark (Proof idea)

The hypothesis holds for *every* integer x , so we may instantiate it at any x we like. Choosing $x = -2$ makes $bx + c$ collapse to $c - 2b$, which is exactly the divisibility claim we want — no other reasoning is needed once $x = -2$ is substituted.

When $x = -2$, we get

$$a \mid (b(-2) + c),$$

that is

$$a \mid (c - 2b).$$

■

Proposition 4.2.10 (Divisibility of Integer Combinations (DIC))

Let $a, b, c \in \mathbb{Z}$. If $a \mid b$ and $a \mid c$, then for all $x, y \in \mathbb{Z}$,

$$a \mid (bx + cy).$$

Remark (Proof idea)

Divisibility means b and c are each an integer multiple of a . Any integer combination $bx + cy$ is then a sum of multiples of a , hence itself a multiple of a — this is exactly what licenses factoring a out below.

Proof

Let $a, b, c \in \mathbb{Z}$. Assume $a \mid b$ and $a \mid c$. Then there exist integers k_1, k_2 such that

$$b = ak_1 \quad \text{and} \quad c = ak_2.$$

Let $x, y \in \mathbb{Z}$ be arbitrary. Then

$$bx + cy = ak_1x + ak_2y = a(k_1x + k_2y).$$

Since $k_1x + k_2y \in \mathbb{Z}$, this shows $a \mid (bx + cy)$. ■

Proposition 4.2.11 (Converse of DIC)

Let $a, b, c \in \mathbb{Z}$. If for all $x, y \in \mathbb{Z}$, $a \mid (bx + cy)$, then $a \mid b$ and $a \mid c$.

Remark (Proof idea)

The hypothesis holds for *every* choice of x, y , so we are free to pick whichever values isolate b and c on their own: $(x, y) = (1, 0)$ leaves $bx + cy = b$, and $(x, y) = (0, 1)$ leaves $bx + cy = c$.

Proof

Let a, b, c be arbitrary integers. Assume that for all $x, y \in \mathbb{Z}$, $a \mid (bx + cy)$.

Taking $x = 1, y = 0$, we get

$$a \mid (b \cdot 1 + c \cdot 0), \quad \text{that is,} \quad a \mid b.$$

Taking $x = 0, y = 1$, we get

$$a \mid (b \cdot 0 + c \cdot 1), \quad \text{that is,} \quad a \mid c.$$

Therefore $a \mid b$ and $a \mid c$. ■

5 Sets

Definition 5.0.1

A *set* is a collection of distinct elements. The *empty set* $\emptyset$ has no elements. For a finite set S , its *cardinality* $|S|$ is its number of elements.

We can describe a set by listing its elements, selecting elements with a property, or generating values:

- Listing: $\{2, 3, 4, 6\}$.
- Selection: $\{n \in \mathbb{Z} : 0 < n < 10 \text{ and } n \text{ is even}\} = \{2, 4, 6, 8\}$.
- Generation: $\{2k : k \in \{1, 2, 3, 4\}\} = \{2, 4, 6, 8\}$.

Example 5.0.2

Order and repetition do not change a set: $\{1, 2, 2\} = \{2, 1\}$. Also, $|\emptyset| = 0$, while $|\{\emptyset\}| = 1$: the latter set contains the empty set as its one element.

5.1 Set-builder notation

Definition 5.1.1

The *selection form* $\{x \in U : P(x)\}$ contains exactly the elements of U for which $P(x)$ is true. The *generation form* $\{f(x) : x \in U\}$ contains the outputs of f , which need not be the inputs.

Example 5.1.2

$\{x^2 : x \in \{-2, -1, 0, 1, 2\}\} = \{0, 1, 4\}$: different inputs may give the same output, which appears only once.

We can combine the forms: $\{f(x) : x \in U, P(x)\}$ first selects the inputs satisfying P , then forms their outputs. For example,

$$\{3a : a \in \mathbb{Z}, 2 \mid a\} = \{6k : k \in \mathbb{Z}\}.$$

5.2 Intervals

For real numbers $a < b$, square brackets include an endpoint and round brackets exclude it:

$$\begin{aligned} [a, b] &= \{x \in \mathbb{R} : a \leq x \leq b\}, & (a, b) &= \{x \in \mathbb{R} : a < x < b\}, \\ [a, b) &= \{x \in \mathbb{R} : a \leq x < b\}, & (a, b] &= \{x \in \mathbb{R} : a < x \leq b\}. \end{aligned}$$

5.3 Set operations

Definition 5.3.1

The *union* $S \cup T$ collects elements in either set; the *intersection* $S \cap T$ keeps

elements in both. The *difference* $S - T$ keeps elements in S but not T :

$$\begin{aligned} S \cup T &= \{x : x \in S \text{ or } x \in T\}, & S \cap T &= \{x : x \in S \text{ and } x \in T\}, \\ S - T &= \{x : x \in S \text{ and } x \notin T\}. \end{aligned}$$

When S is a subset of a universal set U , its *complement* is $\bar{S} = U - S$.

Example 5.3.2

Let $A = [-1, 2)$ and $B = (0, 4]$. Then

$$\begin{aligned} A \cup B &= [-1, 4], & A \cap B &= (0, 2), \\ A - B &= [-1, 0], & B - A &= [2, 4]. \end{aligned}$$

The point 2 belongs to B but not A , so it is excluded from the intersection and included in $B - A$.

Some useful identities, with complements taken relative to U , are

$$\begin{aligned} S \cup \emptyset &= S, & S \cap \emptyset &= \emptyset, \\ S \cup U &= U, & S \cap U &= S, \\ S \cup S &= S, & S \cap S &= S, \\ S \cup \bar{S} &= U, & S \cap \bar{S} &= \emptyset. \end{aligned}$$